# AI Assisted Coding

# Assignment-19.3

**Name: S. Vyshnavi**

**HT.No: 2303A51920**

**Batch-07**

## Lab 19: Assignment – Code Translation Between Programming Languages

### Objective

- Translate code across programming languages.
- Understand syntax, structure, and execution differences.
- Validate correctness and performance of translated code.

### Task 1 – JavaScript to Python

### Prompt Used:

Convert the following JavaScript function into Python using a Pythonic approach:

The function takes an array of numbers, filters values greater than 50, and returns the average of those filtered values.

Ensure:

- Use list comprehension instead of filter()

- Use built-in functions like sum()

- Handle empty lists safely

Also explain differences between JavaScript array methods and Python lists.

### Code:

### JavaScript Code:

main.js

```javascript
function calculateAverage(arr) {
    let filtered = arr.filter(num => num > 50);
    let sum = filtered.reduce((acc, val) => acc + val, 0);
    return filtered.length ? sum / filtered.length : 0;
}

console.log(calculateAverage([40, 60, 80, 30, 90]));
```

Output

```
76.66666666666667

=== Code Execution Successful ===
```

### Python Equivalent:

```python
def calculate_average(arr):
    filtered = [x for x in arr if x > 50]
    return sum(filtered) / len(filtered) if filtered else 0

print(calculate_average([40, 60, 80, 30, 90]))
```

**Output:**

```
76.66666666666667
```

**Differences**

- JavaScript uses .filter() and .reduce().
- Python uses list comprehension + sum().
- Python syntax is more concise and readable

## Task 2 – Python OOP to C++ Class Translation

**Prompt Used:**

Translate the following Python class into an equivalent C++ class:

Requirements:

- Include private attributes
- Implement getter and setter methods
- Include a method that calculates a derived value

Ensure:

- Use proper access specifiers (private, public)
- Use correct data types
- Add constructor implementation

Also explain differences in encapsulation between Python and C++.

**Code:**

**Python Class:**

ass-19.3.py > Student > calculate_grade

```python
class Student:
    def __init__(self, name, marks):
        self.__name = name
        self.__marks = marks

    def get_marks(self):
        return self.__marks

    def set_marks(self, marks):
        self.__marks = marks

    def calculate_grade(self):
        return self.__marks / 10
```

## C++ Equivalent

in.cpp  Share  Run  Output

```cpp
#include <iostream>
using namespace std;

class Student {
private:
    string name;
    int marks;

public:
    Student(string n, int m) {
        name = n;
        marks = m;
    }

    int getMarks() {
        return marks;
    }

    void setMarks(int m) {
        marks = m;
    }

    float calculateGrade() {
        return marks / 10.0;
```

```
8.5

=== Code Execu
```

## Differences

- Python uses __ for private variables.

- C++ uses access specifiers (private/public).
- C++ requires data types and compilation.

## Task 3 – REST API Call Translation (Python → JavaScript)

### Prompt Used:

Convert the following Python API request code into JavaScript using Fetch API:

Requirements:

- Convert GET request

- Parse JSON response

- Handle errors using try-catch

- Use async/await properly

Also explain the difference between Python requests library and JavaScript fetch API.

### Python Code:

## Task 4 – SQL to MongoDB Query Translation

### Prompt used:

Convert the following SQL query into a MongoDB aggregation pipeline:

The query:

- Groups records by category

- Calculates average price

- Filters groups where average price is greater than a given threshold

Also explain differences between SQL and MongoDB aggregation.

## Task 5 – Algorithm Translation (Binary Search)

### Prompt Used:

Implement a binary search algorithm in three programming languages: Python, Java, and JavaScript.

Requirements:

- Maintain the same logic across all implementations

- Handle loops and conditions properly

- Ensure correct input and output handling

Also briefly explain the challenges faced during translation between languages.

## Python:

main.py

```python
def binary_search(arr, target):
    left, right = 0, len(arr)-1
    while left <= right:
        mid = (left + right)//2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1
    return -1

print(binary_search([1,2,3,4,5], 4))
```

Output

```
3

=== Code E
```

## Java:

BinarySearch.java

```java
class BinarySearch {
    public static int search(int arr[], int target) {
        int left = 0, right = arr.length - 1;
        while(left <= right) {
            int mid = (left + right) / 2;
            if(arr[mid] == target)
                return mid;
            else if(arr[mid] < target)
                left = mid + 1;
            else
                right = mid - 1;
        }
        return -1;
    }

    public static void main(String args[]) {
        int arr[] = {1,2,3,4,5};
        System.out.println(search(arr, 4));
    }
}
```

Output

```
3

=== Code Ex
```